

NOTES AND COMMENTS

ESTIMATION OF AVERAGE TREATMENT EFFECTS WITH MISCLASSIFICATION

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This paper considers identification and estimation of the effect of a mismeasured binary regressor in a nonparametric or semiparametric regression, or the conditional average effect of a binary treatment or policy on some outcome where treatment may be misclassified. Failure to account for misclassification is shown to result in attenuation bias in the estimated treatment effect. An identifying assumption that overcomes this bias is the existence of an instrument for the binary regressor that is conditionally independent of the treatment effect. A discrete instrument suffices for nonparametric identification.

KEYWORDS: Binary regressor, program evaluation, treatment effects, misclassification, contamination bias, measurement error, binary choice, binomial response.

1. INTRODUCTION

THIS PAPER PROVIDES CONDITIONS for identification and estimation of the average effect, conditioned on covariates, of a binary treatment, program, or policy on a scalar outcome when treatment may be misclassified. More generally, what is provided is an estimator of the effect of a binary regressor in a conditional mean regression (which may also include other regressors) when the binary regressor is observed with error. This equals the conditional average treatment effect given a weak unconfoundedness assumption.

Misclassification occurs when a binary variable (the treatment indicator) is measured with error, that is, some units are reported to have received treatment when they actually have not, and vice versa. For example, in a returns to schooling analysis the outcome could be wages, the binary variable could be attaining a college degree, and misclassification could arise from school transcript reporting errors. See, e.g., Kane and Rouse (1995) and Kane, Rouse, and Staiger (1999). Bollinger (1996) considers misclassification of union status in wage equations. If treatment is a job training program, misclassification may arise if individuals assigned to the program fail to attend or if those not assigned obtain training elsewhere. Similarly, for medical treatment, individuals assigned a drug might fail to take it and those assigned a placebo might obtain treatment from another source. More generally, misclassification describes any binary variable that is sometimes mismeasured.

This paper first shows that misclassification causes attenuation bias in estimated treatment effects, analogous to the attenuation bias of classically mismeasured variables in linear regression models. Additional assumptions on

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the existence of an instrument are then provided that identify misclassification probabilities and the true probability of treatment (conditional on covariates), in addition to identifying the conditional average treatment effect. Corresponding estimators are variants of Hansen's (1982) generalized method of moments (GMM).

Let Y be an outcome, let X be a vector of covariates, and let T^* be an unobserved true treatment indicator or, more generally, let T^* be any unobserved binary variable that affects Y and can only equal zero or one. Define

$$(1) \quad h^*(X, T^*) = E(Y|X, T^*).$$

Hence $h^*(X, T^*)$ is the conditional mean outcome, given X and treatment T^* . Since T^* is binary, we may without loss of generality rewrite the conditional mean outcome as

$$(2) \quad h^*(X, T^*) = h_0^*(X) + \tau^*(x)T^*,$$

where $h_0^*(X) = h^*(X, 0)$ and

$$(3) \quad \tau^*(x) = h^*(x, 1) - h^*(x, 0).$$

Assume we observe draws of Y , X , and T but not of T^* . The binary variable T is a proxy for T^* that is subject to measurement error. Misclassification occurs when $T \neq T^*$. If T^* were observed instead of T , then $h^*(X, T^*)$ and hence $\tau^*(x)$ would be identified and could be estimated by nonparametric regression.

Das (2004) identifies and estimates $\tilde{h}(X, T) = \tilde{h}_0(X) + \tilde{\tau}(x)T$ in the model $Y = \tilde{h}(X, T) + e$, where $E(e|X) = 0$. Estimation is then based on $E(Y|X) = \tilde{h}_0(X) + \tilde{\tau}(x)E(T|X)$. The Das model allows T to be correlated with e . Similar estimators of models with endogenous binary regressors include Newey and Powell (2003) and Florens and Malavolti (2003). The Das estimator cannot be applied in the present context, because here $E(Y|X) = h_0^*(X) + \tau^*(x)E(T^*|X)$ and, unlike the case with classically mismeasured variables (see, e.g., Aigner (1973)), with a mismeasure T of a binary regressor T^* , $E(T|X) \neq E(T^*|X)$.

Let $X = (Z, V)$. Mahajan (2005) obtains identification of $h^*(x, t)$ assuming that $E(Y|Z, V, T^*) = E(Y|Z, T^*)$ and that $E(T^*|Z, V) \neq E(T^*|Z)$, so V is essentially an instrument that correlates with the true probability of treatment but is conditionally mean independent of the outcome. Hu (2005) obtains identification of $h^*(x, t)$ under similar assumptions for multiple valued discrete treatments t and weakens some of Mahajan's side conditions.

The present paper focuses on identifying and estimating $\tau^*(x)$, the effect on Y of changing T^* . I will refer to $\tau^*(x)$ as the conditional outcome treatment effect, or just the treatment effect, although formally it is just the change in the conditional mean of Y that results from changing the binary variable t^* from

zero to one, holding x fixed, and only equals a true treatment effect when an unconfoundedness condition holds, as discussed later.

Define the mismeasured (treatment) effect $\tau(x) = h(x, 1) - h(x, 0)$ where $h(X, T) = E(Y|X, T)$. Theorem 1 shows that under some general conditions $\tau(x) = m(x)\tau^*(x)$, where $0 \leq m(x) \leq 1$, so mismeasurement of the treatment indicator results in attenuation bias, that is, the mismeasured estimate $\tau(x)$ of the true treatment effect $\tau^*(x)$ is biased toward zero. Related results include Klepper (1988), Manski (1990), Bollinger (1996), and Hotz, Mullin, and Sanders (1997).

Theorem 2 then provides sufficient conditions for identifying and estimating the function $m(x)$ and, hence, the true treatment effect $\tau^*(x)$. As in Mahajan (2005), again let $X = (Z, V)$ and assume $E(T^*|Z, V) \neq E(T^*|Z)$, but replace Mahajan's assumption that $E(Y|Z, V, T^*) = E(Y|Z, T^*)$ with the weaker assumption that $E(Y|Z, V, T^* = 1) - E(Y|Z, V, T^* = 0) = E(Y|Z, T^* = 1) - E(Y|Z, T^* = 0)$. Equivalently, rewriting equation (2) as $h^*(X, T^*) = h_0^*(Z, V) + \tau^*(Z, V)T^*$, Mahajan assumes that both $h_0^*(Z, V)$ and $\tau^*(Z, V)$ do not depend on V , whereas this paper assumes only that $\tau^*(Z, V)$ does not depend on V . For example, a standard regression model where $h^*(x, t)$ is linear in its arguments with nonzero coefficients will satisfy this paper's assumption but violate Mahajan's assumption. Another example is relating wages Y to schooling T^* using a Card (1995, 1996) type distance to school instrument V . The present paper would only require that V be conditionally independent of the increase in wages associated with schooling, rather than conditionally independent of the level of wages. The latter Mahajan requirement could be violated if, e.g., access to schooling is correlated with access to higher paying jobs on average (this example assumes other possible problems associated with distance to schooling such as those described by Carneiro and Heckman (2002) can be ignored).

Nonparametric identification will not require a continuous instrument, so V can be discrete, but it will be required that V take on at least three values. In contrast, Mahajan (2005) requires only a binary valued instrument. Some intuition for this result is that V affects the true probability of treatment, and under Mahajan's assumptions, a change in V affects Y only through this effect on T^* . In contrast, this paper allows V to affect Y both through T^* and through $h_0^*(Z, V)$, so two different changes in V must be observable to separate these effects.

2. RELATING TRUE AND MISMEASURED EFFECTS OF TREATMENT

Let $Y(t)$ denote the outcome from treatment $T^* = t$ for $t = 0, 1$. If

$$(4) \quad E[Y(t)|T^*, X] = E[Y(t)|X],$$

then $\tau^*(x) = E[Y(1) - Y(0)|X = x]$, which is the conditional average treatment effect. Equation (4) is a weak version of the Rubin (1978) or Rosenbaum

and Rubin (1984) unconfoundedness assumption with respect to the true treatment T^* . See, e.g., Heckman, Ichimura, and Todd (1998) for a discussion of this and similar results. In what follows, equation (4) is not imposed. If it does hold where T^* is a treatment indicator, then $\tau^*(x)$ is the conditional average treatment effect; otherwise, $\tau^*(x)$ is just the effect on the mean of Y of changing a binary T^* while holding X fixed.

If T^* were observed without error, then equations (1) and (3) would provide an estimator for $\tau^*(x)$ by replacing expectations with nonparametric regressions. Other existing estimators, e.g., those based on matching or conditional propensity scores, could also be used in that case.

ASSUMPTION A1: *There exists $E(Y|X, T^*, T) = E(Y|X, T^*)$.*

Equivalently, Assumption A1 says that Y is mean independent of $T - T^*$, conditional on X, T^* , so misclassification does not affect the true expected outcome. This is analogous to the classical assumption of independent measurement errors. This can be a substantive assumption if misclassification is due to misperception or deceit on the part of the subject, for example, if T indicates treatment that a respondent believes he or she has received, then Assumption A1 rules out placebo effects. This assumption could also be violated if an individual's propensity to lie about T^* is related to outcomes, e.g., individuals who erroneously claim to have a college degree might also be more aggressive job or wage seekers, in general.

Define $r^*(x)$, $b_0(x)$, $b_1(x)$, and $\tau(x)$ as

$$r^*(x) = E(T^*|X = x) = \Pr(T^* = 1|X = x),$$

$$b_t(x) = \Pr(T = 1 - t|X = x, T^* = t),$$

$$\tau(x) = h(x, 1) - h(x, 0).$$

The function $r^*(x)$ is the conditional (on $X = x$) probability of receiving treatment, while $b_1(x)$ and $b_0(x)$ are the conditional probabilities of misclassifying the treated and the untreated, respectively.

ASSUMPTION A2: *There exist $b_0(x) + b_1(x) < 1$ and $0 < r^*(x) < 1$ for all $x \in \text{supp}(X)$.*

Assumption A2 says first that the sum of misclassification probabilities is less than 1, meaning that, on average, observations of T are more accurate than pure guesses (see, e.g., Bollinger (1996)). In a binomial response model with misclassification, this assumption is what Hausman, Abrevaya, and Scott-Morton (1998) call the monotonicity condition. Without an assumption like this, by symmetry one could never tell if the roles of $t = 0$ and $t = 1$ were reversed, and so, for example, one could not distinguish whether any estimate

of $\tau^*(x)$ corresponded to the effect of changing T^* from zero to one or the negative of this effect. This assumption can be relaxed to $b_0(x) + b_1(x) \neq 1$ if we only wish to identify the magnitude but not the sign of $\tau^*(x)$, which may be useful in applications where the sign of the effect is not in doubt and large misclassification probabilities cannot be ruled out. See Hu (2005) for other possible ways to relax this assumption.

Assumption A2 also requires that, for any value x that we may condition on, there is a nonzero probability of treatment and a nonzero probability of nontreatment, which is needed because a conditional treatment effect cannot be identified if everyone is treated or if no one is treated. If this condition is violated, then we will only obtain identification for values of x where $r^*(x)$ is not zero or one.

Define the functions

$$\begin{aligned} r(x) &= E(T|X = x), \\ h(x, t) &= E(Y|X = x, T = t), \\ \tau(x) &= h(x, 1) - h(x, 0). \end{aligned}$$

Note that $r(x)$ and $\tau(x)$ are the same as $r^*(x)$ and $\tau^*(x)$, except defined in terms of the observed treatment mismeasure T instead of the true treatment T^* , so if treatment were observed without error, then $r(x)$ would be the conditional probability of treatment and $\tau(x)$ would equal the conditional average treatment effect.

THEOREM 1: *If Assumption A1 is satisfied, then there exists a function $m(x)$ with $|m(x)| \leq 1$ such that $\tau(x) = \tau^*(x)m(x)$. If, in addition, Assumption A2 is satisfied, then $m(x) > 0$.*

Proofs are provided in the Appendix. Three expressions for $m(x)$ are provided there. First, $m(x) = \Pr(T^* = 1|X, T = 1) - \Pr(T^* = 1|X, T = 0)$, which shows that $|m(x)| \leq 1$ because m is a difference of probabilities. Second, $[(1 - r(x))r(x)m(x) = [1 - r^*(x)]r^*(x)[1 - b_0(x) - b_1(x)]$, which signs m , and third, $m(x) = M[b_0(x), b_1(x), r(x)]$, where M is defined by

$$(5) \quad m = M(b_0, b_1, r) = \frac{1}{1 - b_1 - b_0} \left(1 - \frac{(1 - b_1)b_0}{r} - \frac{(1 - b_0)b_1}{1 - r} \right),$$

which is used later for point identification and estimation.

Theorem 1 shows that, given just Assumption A1, the magnitude of the mis-measured treatment effect $\tau(x)$ provides a lower bound on the true treatment effect $\tau^*(x)$, and with the addition of Assumption A2, the sign of the mis-measured effect $\tau(x)$ equals the sign of the true effect $\tau^*(x)$. This is like the attenuation bias property of a classically mis-measured regressor coefficient in

a linear bivariate regression, even though here the measurement (misclassification) error is nonclassical and the model is not parameterized.

Theorem 1 also shows that if one is only interested in testing whether $\tau^*(x) = 0$, then misclassification can be ignored, because given Assumptions A1 and A2, $\tau^*(x) = 0$ if and only if $\tau(x) = 0$.

3. IDENTIFICATION

ASSUMPTION A3: *Assume $r(x)$ and $\tau(x)$ are identified.*

Assumption A3 holds given any data set that permits consistent estimation of conditional expectations of observable data. Identification of the conditional outcome function $h(x, t)$ is not required, although an obvious estimator for $\tau(x)$ would be to difference estimates of $h(x, t)$. Identification of $\tau(x)$ provides identification of a lower bound on $\tau^*(x)$ by Theorem 1. Also, if the misclassification probabilities $b_0(x)$ and $b_1(x)$ can be identified, then, given Assumptions A1, A2, and A3, the true treatment effect $\tau^*(x)$ can be identified using $\tau^*(x) = \tau(x)/m(x)$ and equation (5). Additional assumptions will now be made that suffice for this identification.

Partition X into two subvectors V and Z , so $X = (V, Z)$.

ASSUMPTION A4: *For some set $\Omega \subset \text{supp}(V)$, for each $z \in \text{supp}(Z)$, there exists a set $\Omega_z \subset \Omega$ such that for all $v \in \Omega_z$ and $v' \in \Omega_z$, we have $b_t(v, z) = b_t(v', z)$, $\tau^*(v, z) = \tau^*(v', z)$, and for $v \neq v'$, $r^*(v, z) \neq r^*(v', z)$.*

In a small abuse of notation, let $b_t(z)$ and $\tau^*(z)$ denote $b_t(v, z)$ and $\tau^*(v, z)$, respectively, for $v \in \Omega_z$. The distribution of V can be discrete, e.g., V could be a scalar that only takes on a few different values. Assumption A4 says that there exists a variable V that affects the true treatment probabilities r^* , but after conditioning on other covariates does not affect the measurement errors b_t and does not affect τ^* (at least for some values that V might take on). If T^* is a treatment and unconfoundedness holds, then Assumption A4 says that after conditioning on other covariates, V does not affect the conditional average treatment effect, but is correlated with eligibility or selection for treatment.

Having a V that does not affect misclassification probabilities is sometimes used for identification in binomial response models with misclassification. See, e.g., Hausman, Abrevaya, and Scott-Morton (1998), Abrevaya and Hausman (1999), and Lewbel (2000). A typical assumption in misclassified binomial response is that b_0 and b_1 are constants, which would imply that any elements of X could serve as V for that part of Assumption A4.

Having V affect r^* but not τ^* is a weaker version of the type of exclusion assumption that is commonly used in the identification of selection models. See, e.g., Heckman (1990) for a discussion. Close variants of this assumption are used by Manski (1990) to sharpen bounds on treatment effects and by Imbens

and Angrist (1994) to identify local average treatment effects. The τ^* condition is satisfied if $E(Y|Z = z, V = v, T^* = t) = s_1(z, t) + s_2(z, v)$ for some functions s_1 and s_2 .

Assumptions A1, A2, A3, and A4 are all equivalent to (or are implied by) assumptions made by Mahajan (2005), but as discussed in the Introduction, Mahajan also requires that $h^*(v, z, t) = h^*(v', z, t)$ for v and v' as defined in Assumption A4. However, Mahajan only requires that Ω contain two elements, so V can be binary, while the next assumption here requires that Ω contain at least three elements.

ASSUMPTION A5: Each set $\Omega_z \subset \Omega$ contains three elements $v_k \in \Omega_z$, $k = 0, 1, 2$, such that

$$\begin{aligned} & \left(\frac{\tau(v_0, z)}{r(v_1, z)} - \frac{\tau(v_1, z)}{r(v_0, z)} \right) \left(\frac{\tau(v_0, z)}{1 - r(v_2, z)} - \frac{\tau(v_2, z)}{1 - r(v_0, z)} \right) \\ & \neq \left(\frac{\tau(v_0, z)}{r(v_2, z)} - \frac{\tau(v_2, z)}{r(v_0, z)} \right) \left(\frac{\tau(v_0, z)}{1 - r(v_1, z)} - \frac{\tau(v_1, z)}{1 - r(v_0, z)} \right). \end{aligned}$$

The main content of Assumption A5 is that V can take on at least three values. Assumption A5 is expressed in a form that can be empirically tested, because the $\tau(v, z)$ and $r(v, z)$ functions are conditional expectations of observable data, and so can be directly estimated (they are identified by Assumption A3). Assumption A5 can alternatively be written as requiring that $\tau^*(z) \neq 0$, $b_0(z) + b_1(z) \neq 1$, and a certain inequality holds among just the r and r^* functions (see the Appendix for details). Assumption A5 will therefore fail to hold only if the true treatment effect $\tau^*(z)$ is zero or if a complicated nonlinear equality relationship holds among the true and mismeasured conditional treatment probabilities. This would require a perfect coincidence with regard to the levels of these probabilities across all the values V can take on.

If Ω has more than three elements, then Assumption A5 will hold as long as, for each value of z , there exists any one triplet v_0, v_1, v_2 of V values in Ω_z that satisfy the necessary inequality. Note that the triplets v_0, v_1, v_2 are permitted to vary by (i.e., depend on) z .

THEOREM 2: Let Assumptions A1, A2, A3, A4, and A5 hold. Then the conditional misclassification probabilities $b_0(x)$ and $b_1(x)$, the conditional probability of treatment $r^*(x)$, and the effect $\tau^*(x)$ are all identified. Also, if the condition in Assumption A2 that $b_0(x) + b_1(x) < 1$ is replaced by $b_0(x) + b_1(x) \neq 1$, then $\tau^*(x)$ is identified up to sign.

A key component of Theorem 2 is that data on outcomes help to identify misclassification probabilities. In particular, suppressing z for clarity, it follows from Theorem 1 and Assumption A4 that

$$(6) \quad \tau(v_k)M[b_0, b_1, r(v_0)] = \tau(v_0)M[b_0, b_1, r(v_k)].$$

Equation (6) depends only on the identified functions τ and r and on the unknowns b_0 and b_1 . For each z , evaluating this expression at $k = 1$ and $k = 2$ gives two equations in the two unknowns b_0 and b_1 . These equations are non-linear, but the proof of Theorem 2 shows that these equations have a unique solution and thereby identify b_0 and b_1 . Identification of the true treatment effect $\tau^*(x)$ then follows from equation (5).

Equation (6) also shows why identification requires V to take on three values. This equation depends on v_0 and v_k , so evaluating it at $k = 1$ and $k = 2$ requires existence of v_0 , v_1 , and v_2 . Each additional value that V can take on provides another equation that b_0 and b_1 must satisfy, so the larger is the set of values Ω_z , the greater will be the number of overidentifying restrictions that determine b_0 and b_1 at that z .

A binary V would suffice for identification if we had some additional equality restriction on the misclassification probabilities b_0 and b_1 . For example, in some applications it may be known that one or the other of these probabilities is zero, such as when T^* is a job training program where we directly observe everyone who takes the offered program, but we do not always observe when someone who turns down the program gets alternative training elsewhere. An example of a binary V might be a second mismeasured treatment indicator. Theorem 2 could be applied without additional information if we observed three mismeasured treatment indicators, by taking one of them as T and the other two (which together can take on a total of four values) as V .

4. SEMIPARAMETRIC AND NONPARAMETRIC ESTIMATION

To construct estimators, the previous identification conditions will now be expressed in the form of conditional moments. Assume the distribution of V is discrete, define $\Omega = \text{supp}(V) = \{v_0, \dots, v_K\}$, for simplicity assume $\Omega_z = \Omega$, and let $r_k^*(z) = r^*(v_k, z)$. Let $W = (Y, T, V)$.

Define the vector valued function $q_0(z)$ as the vector of $K + 4$ elements

$$(7) \quad q_0(z) = (b_0(z), b_1(z), r_0^*(z), \dots, r_K^*(z), \tau^*(z))$$

and define g as the vector valued function $g[q_0(z), w]$ that consists of the $2K + 2$ elements

$$(8) \quad [b_0(z) + (1 - b_0(z) - b_1(z))r_k^*(z) - T]I(V = v_k) \quad (k = 0, \dots, K),$$

$$(9) \quad \tau^*(z)I(V = v_k) + \frac{YT - (1 - b_1(z))r_k^*(z)\tau^*(z)I(V = v_k)}{b_0(z) + (1 - b_0(z) - b_1(z))r_k^*(z)} \\ - \frac{Y(1 - T) + (1 - b_0(z))(1 - r_k^*(z))\tau^*(z)I(V = v_k)}{1 - [b_0(z) + (1 - b_0(z) - b_1(z))r_k^*(z)]} \\ (k = 0, \dots, K).$$

COROLLARY 1: Define the function q_0 by equation (7) and define the function g as the vector of functions (8) and (9). For any value of z in its support, if Assumptions A1, A2, A3, A4, and A5 hold, then the only function $q(z)$ that satisfies $E[g(q(Z), W)|(Z = z)] = 0$ and has first two elements that are non-negative and sum to less than 1, is $q(z) = q_0(z)$.

The objects we wish to estimate are elements of $q_0(z)$. Corollary 1 shows that the identification based on Theorem 3 can be expressed as the statement that the unknown functions $q_0(z)$ are the solutions to the vector of conditional moments $E[g(q(Z), W)|Z = z] = 0$.

Based on Corollary 1, the functions of interest $q_0(z)$ can be nonparametrically estimated using nonparametric conditional moment GMM based estimators such as Carrasco and Florens (2000), Newey and Powell (2003), or Ai and Chen (2003). In particular, estimation can take the form of parametric conditional moment GMM, replacing the unknown functions with sieve approximations. Empirical likelihood based conditional moment estimation such as Otsu (2003) could also be used. Lewbel (2006a) provides a simple local GMM estimator for this $q_0(z)$.

To construct semiparametric estimators based on Corollary 1, assume we can write $q_0(z) = s(z, \beta_0)$, where s is a known function and β_0 is a finite vector of unknown parameters. In this case, the conditional moments $E[g(q(Z), W)|(Z = z)] = 0$ imply unconditional moments

$$(10) \quad E[\eta_j(Z)g(s(Z, \beta_0), W)] = 0 \quad (j = 1, \dots, J)$$

for bounded functions $\eta_j(Z)$ chosen by the econometrician. Given the unconditional moments of equation (10) for $j = 1, \dots, J$, we may estimate β_0 using Hansen's (1982) generalized method of moments. Asymptotic efficiency can be obtained by using estimated optimal $\eta_j(Z)$ functions as in Newey (1993), Donald, Imbens, and Newey (2003), or Dominguez and Lobato (2004).

With this semiparametric estimator, only the dependence of probabilities and treatment effects on z is parameterized. The dependence of probabilities and treatment effects on v and on unobservables remains nonparametric. Identification of β_0 will depend on the specification of the function s and on $\eta_j(Z)$, but as long as β_0 is identified from $q_0(z) = s(z, \beta_0)$, we can choose $\eta_j(Z)$ functions as previously to identify β_0 . This identification requires that $2K + 2$ (the dimension of s) times J be greater than or equal to the dimension of β_0 .

If Z is discrete with a finite number of support points, then nonparametric estimation of all the functions $q_0(z)$ can be written as a special case of this semiparametric estimator by defining β_0 as the vector of elements $\beta_{j0} = q_0(z_j)$ and letting $\eta_j(Z) = I(Z = z_j)$, where j indexes every value of $z_j = z$ in the support of Z .

It may be of interest to directly estimate the misclassified treatment effect $\tau_k(z) = \tau(v_k, z)$ and the misclassified treatment parameters $r_k(z) = r(v_k, z)$.

By Theorem 1, τ^* is zero if and only if τ is zero, so estimates of misclassified treatment effects could be used to test if the true treatment effects are zero. Estimates of τ and r could also be used in the bounds calculation of Theorem 1 when validity of V as an instrument is in doubt, and they can be used to test Assumption A5. To express the misclassified functions as conditional moments, define $Q_0(z)$ as the $2K + 2$ vector

$$(11) \quad Q_0(z) = (r_0(z), \dots, r_K(z), \tau_0(z), \dots, \tau_K(z))$$

and define $G[Q_0(z), w]$ as the vector valued function that consists of the $2K + 2$ elements

$$(12) \quad [r_k(z) - T]I(V = v_k) \quad (k = 0, \dots, K),$$

$$(13) \quad \left(\frac{YT}{r_k(z)} - \frac{Y(1-T)}{1-r_k(z)} - \tau_k(z) \right) I(V = v_k) \quad (k = 0, \dots, K).$$

COROLLARY 2: Define the function Q_0 by equation (11) and define the function G as the vector of functions (12) and (13). For any value of z in its support, if Assumptions A1 and A2 hold, then the only function $Q(z)$ that satisfies the equations $E[G(Q(Z), W)|(Z = z)] = 0$ is $Q(z) = Q_0(z)$.

Nonparametric or semiparametric estimation based on Corollary 2 proceeds exactly like estimation based on Corollary 1 as previously described, replacing q and g with Q and G .

5. CONCLUSIONS

This paper provides bounds and conditions for nonparametric point identification of the effect on an outcome of changing a possibly mismeasured binary variable. Given a weak form of the unconfounding assumption, this provides identification of conditional average treatment effects when the treatment indicator may be mismeasured. Estimators that employ these identification conditions are provided, based on direct estimation of relevant conditional expectations.

Lewbel (2006a) and an addendum to this paper (Lewbel (2006b)) describe both semiparametric and local GMM nonparametric estimation of a vector of functions $q(Z)$ based on moments of the form $E[g(q(Z), W)|(Z = z)] = 0$ (Corollaries 1 and 2 are examples of such functions and moments), and provide both a small Monte Carlo and an empirical application of Theorem 2 and Corollaries 1 and 2, relating wages to attainment of an undergraduate degree, allowing for misclassification errors in transcript reports.

It would be useful to explore how other binary covariate effect estimators such as matching and propensity score based methods might be adapted to the

present application where the binary covariate such as treatment is mismeasured.

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APPENDIX

Define $p_t(X) = E(T^*|X, T = t) = \Pr(T^* = 1|X, T = t)$. Suppressing the X argument for clarity, some relationships to be used later are, by Bayes theorem,

$$(14) \quad p_0 = \frac{b_1 r^*}{1 - r} \quad \text{and} \quad p_1 = \frac{(1 - b_1) r^*}{r}.$$

Also,

$$(15) \quad r = E(T) = E(T|T^* = 1) \Pr(T^* = 1) + E(T|T^* = 0) \Pr(T^* = 0) \\ = (1 - b_1) r^* + b_0 (1 - r^*),$$

which gives $r = b_0$ when $b_0 + b_1 = 1$; otherwise

$$(16) \quad r^* = \frac{r - b_0}{1 - b_0 - b_1} \quad \text{and} \quad 1 - r^* = \frac{1 - b_1 - r}{1 - b_0 - b_1}.$$

PROOF OF THEOREM 1: Continuing to suppress the X argument, we have by equations (1) and (2) and Assumption A1 that $E(Y|T^*, T) = h_0^* + \tau^* T^*$. By the law of iterated expectations, this gives $E(Y|T = t) = h_0^* + \tau^* p_t$. Then whereas $\tau = E(Y|T = 1) - E(Y|T = 0)$, we obtain $\tau = (p_1 - p_0)\tau^*$, so m in Theorem 1 equals $p_1 - p_0$ and $-1 \leq m \leq 1$ follows from m equaling the difference between two probabilities.

Using equation (14),

$$(17) \quad m = p_1 - p_0 = \frac{(1 - b_1) r^*}{r} - \frac{b_1 r^*}{1 - r} \\ = \frac{r^*}{(1 - r)r} (1 - b_1 - r)$$

and using equation (16) for $1 - r^*$,

$$(18) \quad (1 - r)rm = (1 - r^*)r^*(1 - b_0 - b_1).$$

Because probabilities r and r^* lie between 0 and 1, this shows that $m > 0$ when Assumption A2 holds, $m = 0$ when $b_0 + b_1 = 1$, and $m < 0$ when the sum of the

misclassification probabilities $b_0 + b_1$ is greater than 1. Also, substituting equation (16) for r^* into equation (17) yields, after some algebraic simplification, equation (5). *Q.E.D.*

PROOF OF THEOREM 2: For clarity suppress z , and for the functions r and τ denote $r_k = r(v_k)$ and $\tau_k = \tau(v_k)$. For any given z , we have, for all $v_0 \in \Omega_z$ and $v_k \in \Omega_z$, that, by Theorem 1 and equation (5),

$$\begin{aligned} 0 &= M(b_0, b_1, r_k)\tau_0 - M(b_0, b_1, r_0)\tau_k, \\ (19) \quad 0 &= \left(1 + \frac{(b_1 - 1)b_0}{r_k} + \frac{(b_0 - 1)b_1}{1 - r_k}\right)\tau_0 \\ &\quad - \left(1 + \frac{(b_1 - 1)b_0}{r_0} + \frac{(b_0 - 1)b_1}{1 - r_0}\right)\tau_k, \end{aligned}$$

$$(20) \quad 0 = (1 - b_1)b_0\left(\frac{\tau_0}{r_k} - \frac{\tau_k}{r_0}\right) + (1 - b_0)b_1\left(\frac{\tau_0}{1 - r_k} - \frac{\tau_k}{1 - r_0}\right) + \tau_k - \tau_0.$$

Write this equation as

$$0 = B_0w_{0k} + B_1w_{1k} + w_{2k},$$

where $B_t = (1 - b_{1-t})b_t$ and each w_{jk} is a function of r_0, r_k, τ_0 , and τ_k . Given that $\Omega_z \in \Omega$ contains three elements v_0, v_1 , and v_2 , we have two equations $0 = B_0w_{0k} + B_1w_{1k} + w_{2k}$ for $k = 1, 2$ that are linear in the two unknowns B_0 and B_1 , and so can be uniquely solved as long as the matrix of elements w_{jk} , $j = 0, 1, k = 1, 2$, is nonsingular. The determinant of this matrix is

$$(21) \quad \left(\frac{\tau_0}{r_1} - \frac{\tau_1}{r_0}\right)\left(\frac{\tau_0}{1 - r_2} - \frac{\tau_2}{1 - r_0}\right) - \left(\frac{\tau_0}{r_2} - \frac{\tau_2}{r_0}\right)\left(\frac{\tau_0}{1 - r_1} - \frac{\tau_1}{1 - r_0}\right)$$

and the inequality in Assumption A5 makes this determinant nonzero, as required.

Now let $s = 1 - b_1 - b_0$. It follows from $B_t = (1 - b_{1-t})b_t$ that $(s + b_0)b_0 = B_0$ and $2b_0 = B_0 - B_1 + 1 - s$. Substituting the second of these equations into the first and solving for s gives

$$1 - b_1 - b_0 = s = \pm[(B_0 - B_1 + 1)^2 - 4B_0]^{1/2}$$

if the assumption regarding s is $s \neq 0$. Then we have that s is identified up to sign. By Theorem 1, $\tau^* = \tau(v)/M[b_0, b_1, r(v)]$ and

$$M[b_0, b_1, r(v)] = \frac{1}{s} \left(1 - \frac{B_0}{r(v)} - \frac{B_1}{1 - r(v)}\right),$$

so it follows that τ^* is identified up to sign. Making the stronger assumption that $s > 0$, we have s is identified, so b_0 and b_1 are now identified by

$b_0 = (B_0 - B_1 + 1 - s)/2$ and $b_1 = -(B_0 - B_1 + 1 + s)/2$. In addition by equations (16) and (5), and Theorem 1, identification of these misclassification probabilities means that r^* and τ^* are also identified.

Define $R_k = [(1 - r_k^*)r_k^*]/[(1 - r_k)r_k]$. Using $\tau = m\tau^*$ and equation (17), the determinant (21) can be rewritten as

$$\left[\left(\frac{R_0}{r_1} - \frac{R_1}{r_0} \right) \left(\frac{R_0}{1 - r_2} - \frac{R_2}{1 - r_0} \right) - \left(\frac{R_0}{r_2} - \frac{R_2}{r_0} \right) \left(\frac{R_0}{1 - r_1} - \frac{R_1}{1 - r_0} \right) \right] \\ \times (1 - b_0 - b_1)\tau^*.$$

So the nonzero determinant condition can equivalently be written as requiring that $\tau^* \neq 0$, $b_0 + b_1 \neq 1$, and the previously described function of r_k^* and r_k for $k = 0, 1, 2$ not equal to zero. Q.E.D.

PROOF OF COROLLARY 1: To ease notation, drop the argument z everywhere and let all subsequent expectations be conditional on $Z = z$. Let $I_k = I(V = v_k)$. Having the mean of equation (8) equal zero makes $b_0 + (1 - b_0 - b_1)r_k^* = E(I_k T)/E(I_k)$, which equals the true r_k by definition of r_k . Solving the resulting equation $b_0 + (1 - b_0 - b_1)r_k^* = r_k$ for r_k^* and substituting the result into equation (9) gives

$$\left(\frac{YT}{r_k} - \frac{(1 - b_1)\tau^*}{r_k} \frac{r_k - b_0}{1 - b_0 - b_1} - \frac{Y(1 - T)}{1 - r_k} - \frac{(1 - b_0)\tau^*}{1 - r_k} \frac{1 - b_1 - r_k}{1 - b_0 - b_1} + \tau^* \right) I_k.$$

Setting the mean of this result to zero and dividing by $E(I_k)$ gives

$$\frac{E(YTI_k)}{r_k E(I_k)} - \frac{(1 - b_1)\tau^*}{r_k} \frac{r_k - b_0}{1 - b_0 - b_1} - \frac{E[Y(1 - T)I_k]}{(1 - r_k)E(I_k)} - \frac{(1 - b_0)\tau^*}{1 - r_k} \frac{1 - b_1 - r_k}{1 - b_0 - b_1} + \tau^* = 0,$$

which, using $r_k = E(TI_k)/E(I_k)$, simplifies to

$$\frac{E(YTI_k)}{E(TI_k)} - \frac{(1 - b_1)\tau^*}{r_k} \frac{r_k - b_0}{1 - b_0 - b_1} - \frac{E[Y(1 - T)I_k]}{E[(1 - T)I_k]} - \frac{(1 - b_0)\tau^*}{1 - r_k} \frac{1 - b_1 - r_k}{1 - b_0 - b_1} + \tau^* = 0,$$

which, after rearranging terms and using $E(TI_k) = \text{prob}(T = 1, V = v_k)$,

gives

$$E(Y|T=1, V=v_k) - E(Y|T=0, V=v_k) \\ = \left(\frac{1-b_1}{r_k} \frac{r_k-b_0}{1-b_0-b_1} + \frac{1-b_0}{1-r_k} \frac{1-b_1-r_k}{1-b_0-b_1} - 1 \right) \tau^*,$$

which, by the definitions of the functions τ and m , reduces to $\tau(v_k) = M(b_0, b_1, r_k)\tau^*$. It has now been shown that the conditional mean of g equaling zero is equivalent to $r(v_k) = b_0 + (1-b_0-b_1)r_k^*$ and $\tau(v_k) = M[b_0, b_1, r(v_k)]\tau^*$ with the true functions $r(v_k)$ and $\tau(v_k)$, and, by Theorem 2, the only solutions to these equations for $k=0, \dots, K$ that also satisfy $b_0 \geq 0$, $b_1 \geq 0$, and $b_0 + b_1 < 1$ are the true values of $r_0^*, \dots, r_K^*, b_0, b_1$, and τ^* . Q.E.D.

PROOF OF COROLLARY 2: Setting the conditional mean of equation (12) equal to zero and solving for $r_k(z)$ yields the definition of $r_k(z)$, and setting the conditional mean of equation (13) equal to zero and solving for $\tau_k(z)$ yields the definition of $\tau_k(z)$. Q.E.D.

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